

DLBDS00FPP01 Object Oriented and Functional Programming with Python

HABIT TRACKER

Development & Reflection Phase

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1. Introduction

➤ What this project does:

1 Efficient Habit Tracking

Allows users to define, complete and monitor habits easily

2 Python Classes

Implements Python classes for structured and reusable code

3 Data Storage

Progress is saved in JSON files, keeping data persistent and accesible

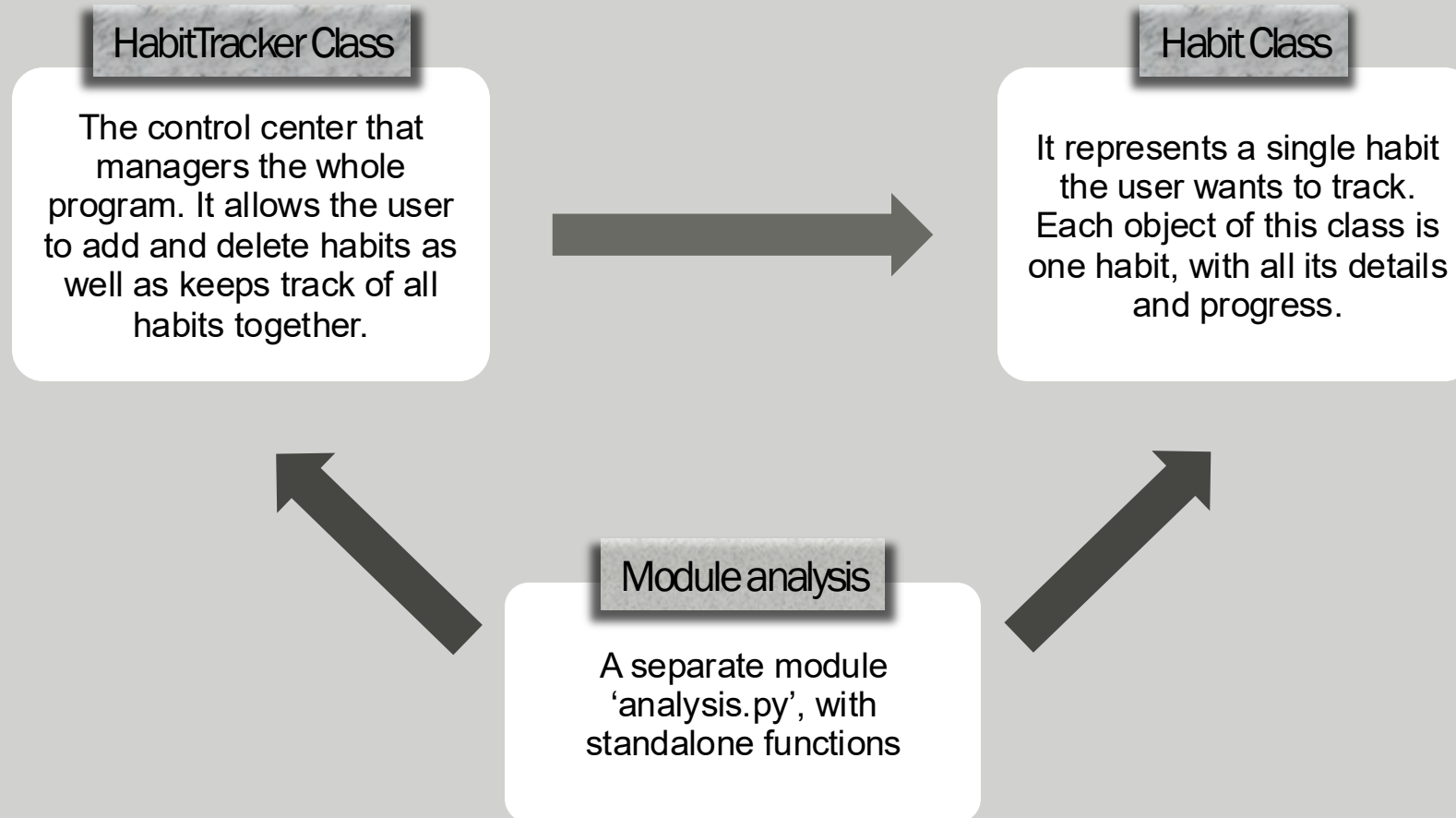
4 Interactive CLI

Users interact through a command-line menu to manage habits quickly

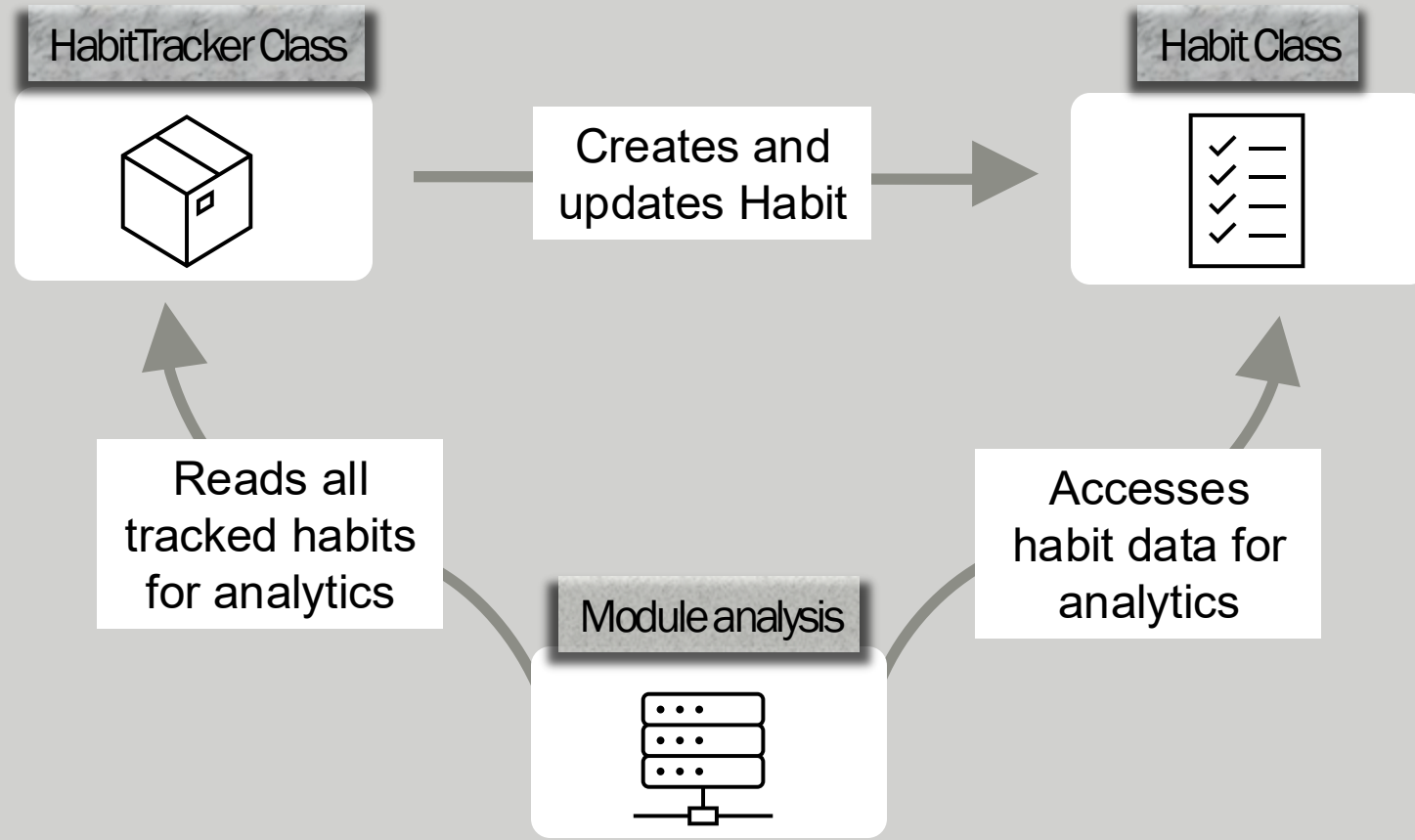
5 Progress Analysis

Provides insights on streak, identifies habit consistency and filters habits by daily or weekly periodicity

2. System Design



3. Data Flow



4. Tools and Implementations



Python 3.7+ → Modern version for Object-Oriented and functional programming



JSON Storage → Keeps habits and completions persistent



Command-Line Interface → Users interact via menu to create, complete, delete and view habits



Code Documentation → Docstrings and comments for clarity and maintainability

5. How to use the app:

Creation of a Habit

`python main.py`

```
=====
HABIT TRACKER - MAIN MENU
=====
1. Start (Load Data)
2. Add habit
3. Remove habit
4. Complete habit
5. View habit
6. List all habits
7. List habits by periodicity
8. View analysis
9. Load test data
10. Exit
=====
Select an option (1-9): █
```

```
--- ADD NEW HABIT ---
Enter habit name: Go to the gym
Enter periodicity (daily/weekly): daily
Habit 'Go to the gym' added with ID 1.
Habits saved to habits.json.
```

Run main.py to open the CLI menú.

Choose option 'Add habit' in the menu.

Enter habit name.

Select daily or weekly periodicity.

Habit is saved and visible in the habit list.

6. How to use the app:

Completion of a Habit

```
=====
HABIT TRACKER - MAIN MENU
=====
1. Start (Load Data)
2. Add habit
3. Remove habit
4. Complete habit
5. View habit
6. List all habits
7. List habits by periodicity
8. View analysis
9. Load test data
10. Exit
=====
Select an option (1-9): █
```

↓

```
--- ALL HABITS ---
ID: 1 | Name: Go to the gym | Periodicity: daily | Streak: 0
Enter the ID of the habit to complete: 1
Habit 'Go to the gym' marked as completed for today.
Habits saved to habits.json.
```

Choose option 'Complete habit' from the menu.

Enter the ID of the habit to mark as done for today (daily) or this week (weekly).

Completion is recorded in JSON file for persistence.

7. How to use the app:

Analysis Habits

```
=====
HABIT TRACKER - MAIN MENU
=====
1. Start (Load Data)
2. Add habit
3. Remove habit
4. Complete habit
5. View habit
6. List all habits
7. List habits by periodicity
8. View analysis
9. Load test data
10. Exit
=====
Select an option (1-9): █
```

↓

```
--- ANALYSIS MENU ---
a. View current streak for a habit
b. View longest streak
c. View lowest streak
d. Back to main menu
Select an option (a-c): █
```

Choose option 'View analysis'.

Select options:

- a. View current streak for a habit
- b. View longest streak
- c. View lowest streak
- d. Back to main menu

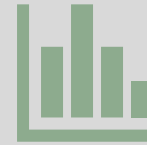
8. Conclusion



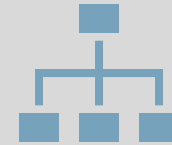
Implemented a modular habit tracker using Python and OOP



Data stored persistently with JSON for reliability



Users can easily create, manage and analyse habits



Project shows the importance of structured design and reusability